



Carbon prices in Alberta's Power Market

And a few other things we're working on...

Theme 6: Grids and Storage

Presentation by Andrew Leach

Storage content provided by Marc Secanell

Smart Grid content provided by Ryan Li

Future Energy Systems Symposium

University of Alberta

May, 2019



Grids and Storage Theme Summary

Goal

Develop advanced smart grid and energy storage technologies and study the economics and policy to enable the high penetration of renewable energy into the grid.

Active projects: Smart Grids

- Future Smart Grid Structures
- Operational Decision Support for Smart Grid
- Energy Management Strategies for Distributed Energy Resources
- Surface Science Hub for Clean Technology



Grids and Storage Theme Summary

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Develop advanced smart grid and energy storage technologies and study the economics and policy to enable the high penetration of renewable energy into the grid.

Active projects: Storage

- Distributed Energy Storage
- Utility-Scale Energy Storage
- Next Generation Energy Storage Using Unconventional Materials
- Research on Interfacial Control of Solid State Lithium Batteries



Grids and Storage Theme Summary

Goal

Develop advanced smart grid and energy storage technologies and study the economics and policy to enable the high penetration of renewable energy into the grid.

Active projects: Policies & Markets

- Economic policy and the future electricity grid
- Carbon pricing and other policies impacts on the market for storage
- The complementary roles of renewables and storage on the grid
- Computing Platform for Managing Transactive Microgrids



Carbon Pricing, Electricity Markets and Output-Based Allocations

- Impacts of carbon pricing has become a central policy question in Canada and globally
- Electricity markets, at least in certain jurisdictions are ideal environments in which to study responses to carbon prices
- Rich, hourly electricity market data allows us to observe firm-level decisions with respect to carbon price pass through
- Fabra and Reguant (2014) shows how firms respond to emissions credit prices and that lump-sum credit allocations were not distortionary in the short run
- Brown et al. (2018) looks at Alberta policy changes which alter both carbon prices and output-based credit allocations



Alberta's Power Market is an Ideal Laboratory

- Alberta's power market is relatively small, isolated, and features a mix of generation technologies
- Our peak internal load for 2018 was 11,697 MW, while our lowest observed internal load was 7,819 MW
- We have relatively minimal inertia capacity. Alberta is connected to WECC (1325 MW export and 1500 MW import path-rating) via Montana and BC) and to Saskatchewan (150 MW path-rating) but significant hourly variability in capacity which helps us with identification
- Supply mix (2018) was 44.8% coal, 21.1% net-to-grid from combined heat and power plants, 18.5% natural gas, 2.9% hydro, 6.4% wind, 1.0% other sources including solar and 5.2% imports.



Alberta's Policy Changes as Treatments

- From 2007 to 2015, Alberta's *Specified Gas Emitters Regulation* imposed a carbon price of \$15 per tonne with output-based allocations at a rate of 88% of a facility's historic emissions intensity if the facility's annual emissions were more than 100,000 tonnes per year
- In 2015, these parameters were changed to \$20 per tonne with allocations equal to 85% of historic emissions intensities in 2016 and \$30 per tonne with allocations of 80% in 2017.
- 2007-2017 system also had generous emissions credits for combined heat and power systems
- In 2018, the system changed again, to one with output-based allocations fixed for all generators (0.37t/MWh) including for net generation from combined heat and power.



Data Richness

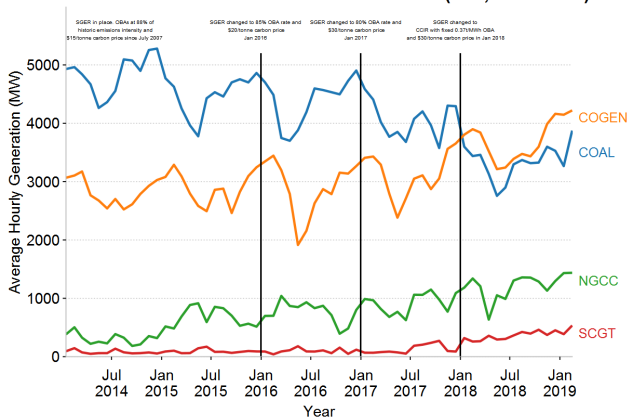
We have data on:

- Hourly merit order (power offers) by unit with data on who held offer control for that unit at that time
- Hourly forecast and actual loads and prices
- Hourly weather for Edmonton, Calgary, and Fort McMurray
- Hourly imports, exports, and intertie capabilities
- Hourly wind forecasts including 3 and 7 day advanced forecasts.
- Plant-level emissions intensities and historic allocation rates under *SGER*



Carbon Pricing Policy and Power Generation Sources

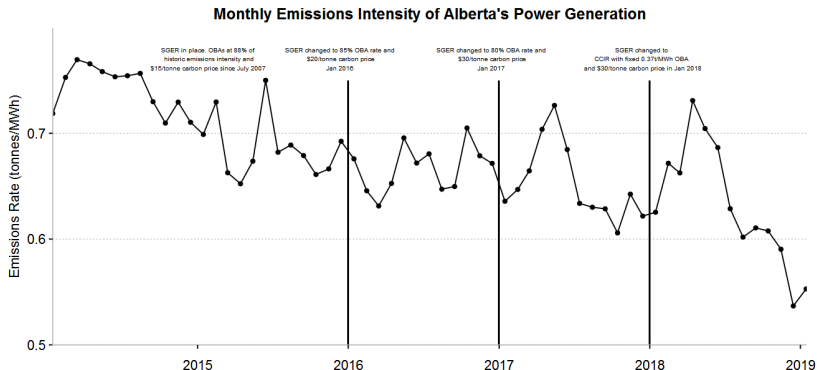
Coal and Gas Generation and Carbon Prices (MW, 2014-2018)



Source: AESO Data, accessed via NRGStream



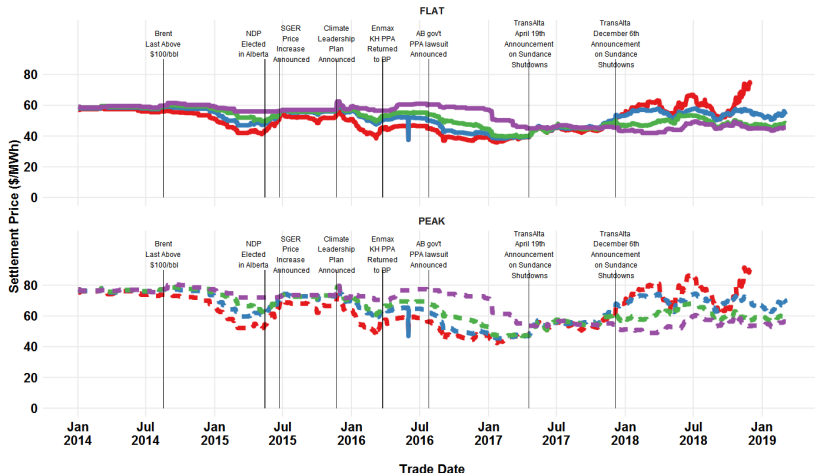
Carbon Intensity of Alberta Power





Alberta Power Prices and Carbon Policy

Alberta Power Forward Contract Values (Calendar Strip)





Secanell group

- Design, procurement and installation of low and high temperature electrolysis fabrication and testing facilities
- Fabrication of single-cell low temperature electrolyzer prototypes using iridium and iridium oxide catalysts
- Demonstrated, for the first time, the use of inkjet printing as a feasible technology to fabricate low temperature electrolyzer cells (<https://doi.org/10.1149/2.1101807jes>)

*Journal of The Electrochemical Society*, **165** (7) F543-F552 (2018)

Analysis of Inkjet Printed Catalyst Coated Membranes for Polymer Electrolyte Electrolyzers

Manas Mandal,^{*} Antoni Valls, Niklas Gangnus, and Marc Secanell^{*,*,z}*Energy Systems Design Laboratory, Department of Mechanical Engineering, University of Alberta, Edmonton, Alberta T6G 1H9, Canada*

Inkjet printing (IJP) is studied as a novel fabrication method for catalyst coated membranes (CCMs) for polymer electrolyte electrolyzers. IrO₂ and Pt/C inks were deposited by IJP over the membrane to fabricate anode and cathode electrodes, respectively. Optical microscopy, scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDX) were used for ex-situ surface and thickness characterization of the anode catalyst layer (CL). SEM images show the CL is uniform and well adhered to the membrane. EDX images show even distribution of the catalyst and ionomer in the CL. Hydrogen cross-over, cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) tests were performed to estimate hydrogen cross-over, double layer capacitance (C_{dl}) and cell resistances. Cross-over results show that the membrane is not damaged during printing. A kinetic study revealed Tafel slopes similar to those in literature. Electrochemical performance tests showed that inkjet printed electrolyzer CCMs achieved 1 and 2 A/cm² current densities at average potentials of 1609 mV and 1696 mV (NRE211) and average of 1743 mV and 1977 mV (N117) respectively. The electrolyzer performance improved slightly when the printing piezo-voltage decreased but with a cost of higher fabrication time. The stability of the electrode was inline with literature data. The proposed electrodes outperform most of the previously reported electrolyzer data in the literature using commercial IrO₂ catalyst.

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Manuscript submitted January 25, 2018; revised manuscript received April 16, 2018. Published May 19, 2018. This was Paper 1665 presented at the National Harbor, Maryland Meeting of the Society, October 1–5, 2017.



Secanell Group

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- Fabrication of single-cell low temperature electrolyzer prototypes using iridium and iridium oxide catalysts
- Demonstrated, for the first time, the use of inkjet printing as a feasible technology to fabricate low temperature electrolyzer cells (<https://doi.org/10.1149/2.1101807jes>)
- Development of a novel nano-pompon-like iridium catalyst superstructure for low temperature fuel cell electrolyzers (<https://doi.org/10.1016/j.jcat.2019.01.018>)



Secanell Group



Journal of Catalysis
Volume 371, March 2019, Pages 57-70



Decoupling structure-sensitive deactivation mechanisms of Ir/IrO_x electrocatalysts toward oxygen evolution reaction

Xuehai Tan ^{a, b}, Jing Shen ^a, Natalia Semagina ^{a, R, B}, Marc Secanell ^{b, R, B}

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<https://doi.org/10.1016/j.jcat.2019.01.018>

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Highlights

- Ir/IrO_x stability in OER was studied at identical electro-oxidative conditions.
- Deactivation paths were connected to the Ir/IrO_x chemical and structural properties.
- The loss of intrinsic activity and ESCA were decoupled.
- The OER activity regeneration of Ir/IrO_x was demonstrated.



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- Development of novel micro-structures for high temperature solid oxide electrolyzers and fuel cells (<https://doi.org/10.1016/j.electacta.2018.02.055>)



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Electrochimica Acta

Volume 268, 1 April 2018, Pages 195-201



The effect of pore-former morphology on the electrochemical performance of solid oxide fuel cells under combined fuel cell and electrolysis modes

Miguel A. Laguna-Bercero ^a , Amir R. Hanifi ^b, Lucile Menand ^c, Navjot K. Sandhu ^b, Neil E. Anderson ^b, Thomas H. Etsell ^b, Partha Sarkar ^d



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HQP include 5 PhD students and 1 MSc. 5 faculty (1 in MECE, 3 in CHEME



Smart Grids

Most renewable energy based power generation, energy storage systems and modern electronic loads are based on DC technologies. This necessitates new grid structure such as hybrid AC/DC and DC grids

Main objectives

- Development of hybrid AC-DC grids technologies
- over electronics technologies, and power management technology

HQP include PI: Ryan Li, co-PI: Khajehoddin, Kish, Mohamed, Salmon, Liang. 18 total HQP since 2017



Main progress so far

- New DC/AC and DC/DC converter technologies for future grids
- Power management, power quality management
- Over 60 publications and 8 new IP

Collaborators

- NSERC CREATE initiative with RWTH Aachen
- Collaboration with Tsinghua Univ
- Alberta utility companies
- Manufacturers: Rockwell Automation, Landmark,
- Alberta CTD projects on "Smart Gas Field Microgrids" and "Next-Gen Modular Green Building Energy Systems With Optional Energy Storage"

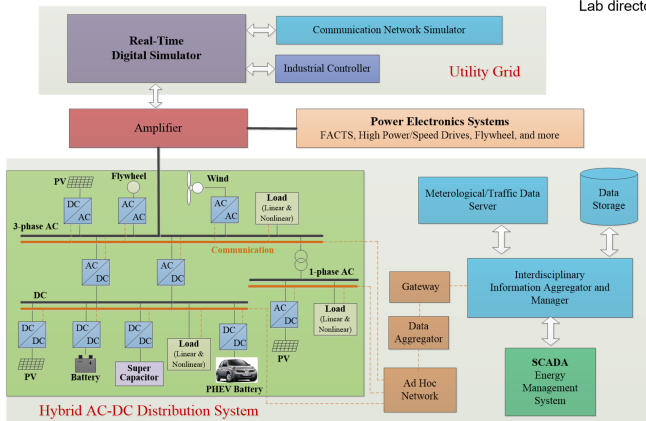


Smart Grid Lab

Future AC-DC Smart Grid Technologies Lab (T06-P01)

Under renovation. To be ready in Q4 2019

Lab director: Ryan Li



Enabled Smart Grid Research:

Smart and active distribution systems

Hybrid AC/DC system control, operation, and protection

Cyber security of smart grids

Power electronics modelling and design



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